

Multi-Phase Fine-Tuning of EfficientNet for Skin Lesion Classification with Comparative Analysis

CN7023- Artificial Intelligence & Machine Vision
Atika AhmedSudi

Data Description

- Dataset Name: Skin Cancer MNIST: HAM10000
- Dataset Link: <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>
- Dataset Size: 2.9 GB
- Image Dimensions: 28x28 coloured images and grayscale images,
- Number of Classes:7
- Number of Images per Class:

Class 1 - Melanocytic Nevi: 6,705

Class 2 - Melanoma: 1,113

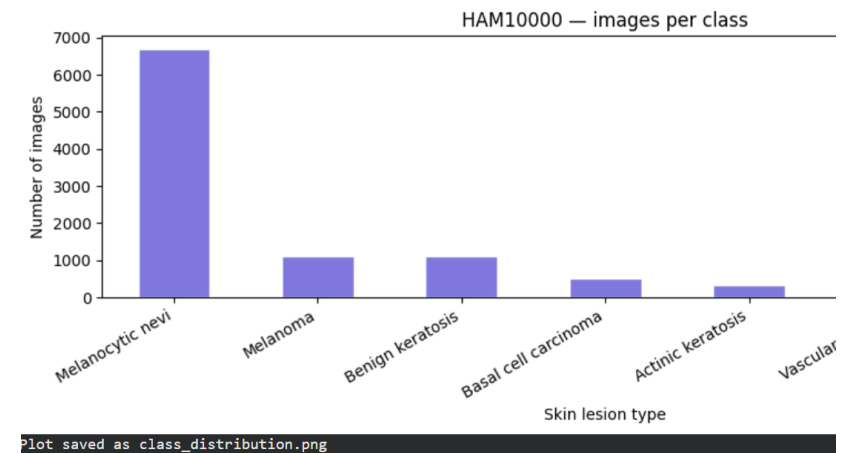
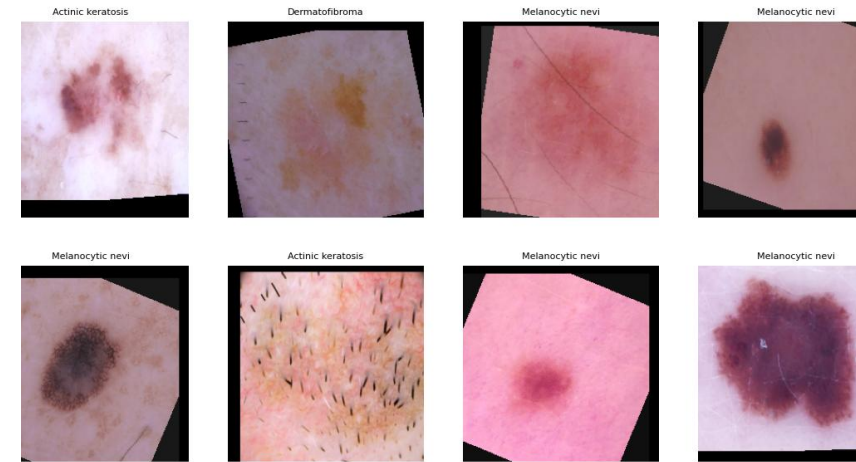
Class 3 - Benign Keratosis: 1,099

Class 4 - Basal Cell Carcinoma: 514

Class 5 - Actinic Keratoses: 327

Class 6 - Vascular lesions: 142

Class 7 - Dermatofibroma: 115



Data Pre-processing & Encoding

- **Image Standardization:** All images resized to **224x224 pixels** using bilinear interpolation for compatibility with EfficientNet-B0.
- **Label Encoding:** Diagnostic categories mapped to integer indices
- **Normalization:** Pixel values normalized using standard ImageNet statistics: Mean [0.485, 0.456, 0.406] and Std [0.229, 0.224, 0.225].
- **Augmentation Pipeline:**
 - **Geometric:** Random horizontal/vertical flips and rotations (15°).
 - **Photometric:** Color jittering (brightness, contrast, and saturation by 0.2) and random affine translations.
 - **Advanced: MixUp augmentation** applied during the final training phase to enhance robustness.

Network Architecture of EfficientNET B0

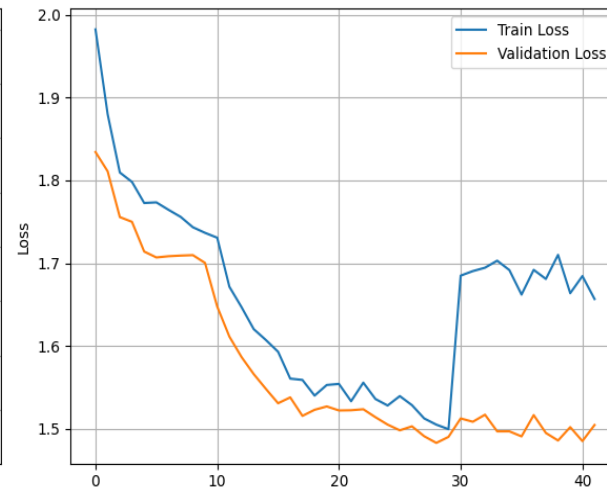
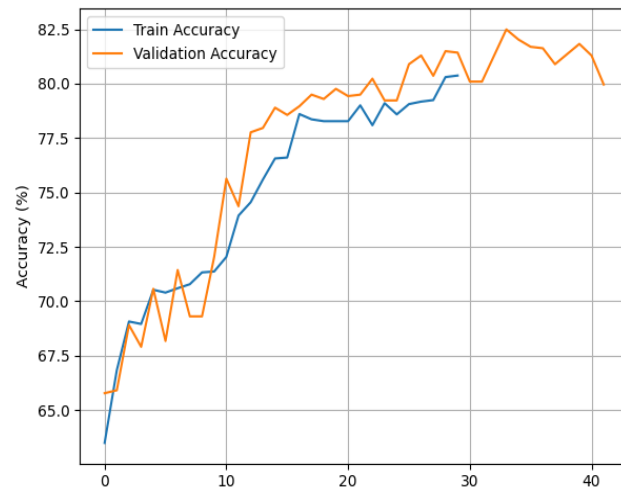


Network Training Options

- **Loss Function:** Categorical Cross-Entropy with **clipped class weights** (min 0.5, max 5.0) and **label smoothing (0.1)** to handle imbalance.
- **Staged Training Strategy:**
 - **Phase 1 (Head Only):** Base frozen; 10 epochs; Adam (LR=1e-3).
 - **Phase 2 (Top Unfrozen):** Last 4 MBConv blocks unfrozen; 20 epochs; Differential LR (Head: 1e-4, Base: 3e-5); CosineAnnealingLR scheduler.
 - **Phase 3 (Full Fine-Tuning):** All layers unfrozen; 10 epochs; Adam (LR=1e-5); OneCycleLR scheduler.
- **Optimization:** Batch size of 32; **ReduceLROnPlateau** scheduler monitoring validation loss in initial phases.

Results

- Train & Validation Accuracy
- Confusion matrix



Confusion Matrix (Test Set)

	Actinic keratosis	Basal cell carcinoma	Benign keratosis	Dermatofibroma	Melanoma	Melanocytic nevi	Vascular lesions
Actinic keratosis	34	5	6	0	3	1	0
Basal cell carcinoma	2	62	6	0	0	5	2
Benign keratosis	11	9	108	0	18	19	0
Dermatofibroma	1	1	1	10	1	3	0
Melanoma	10	3	17	0	97	37	3
Melanocytic nevi	5	14	28	2	44	906	7
Vascular lesions	0	2	0	0	1	1	18

Predicted Label

Comparative analysis

precision	recall	f1-score	support		
Actinic keratosis		0.54	0.69	0.61	49
Basal cell carcinoma		0.65	0.81	0.72	77
Benign keratosis		0.65	0.65	0.65	165
Dermatofibroma		0.83	0.59	0.69	17
Melanoma		0.59	0.58	0.59	167
Melanocytic nevi		0.93	0.90	0.92	1006
Vascular lesions		0.60	0.82	0.69	22
	accuracy			0.82	1503
	macro avg	0.68	0.72	0.69	1503
	weighted avg	0.83	0.82	0.82	1503



Limitation, Future work & Conclusion

- **Limitations:** Class imbalance; **clinical performance gap** in melanoma sensitivity; lack of **external validation** on non-Austrian datasets(Haenssle, H.A. *et al.* (2018))
- **Future Work:**
 - Implement **Focal Loss** to focus on "hard" examples.
- **Conclusion:** EfficientNet-B0 transfer learning is a highly viable approach for automated triage, but refinement in **minority class sensitivity** is essential before real-world clinical deployment.

References

- Kaggle, (2018) <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000> (Accessed: 29 April 2026).
- Haenssle, H.A. et al. (2018) 'Man against machine: diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists', *Annals of Oncology*, 29(8), pp. 1836–1842. doi:10.1093/annonc/mdy166.
- Das, S. and Addya, R.K. (2025) 'Skin cancer detection and classification through medical image analysis using efficientnet', *NDT*, 3(4), p. 23. doi:10.3390/ndt3040023.